

Module 2

Transient Analysis of First and Second-Order Circuits

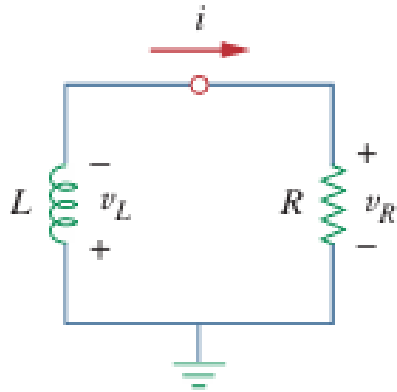
Dr. Elizabeth Rufus

Professor, School of Electronics Engineering

The Source-Free RL Circuit

Consider the series connection of a resistor and an inductor.

The aim is to determine the circuit response, which we will assume to be the current through the inductor



Consider a series combination of a resistor and an inductor.

Inductor current cannot change instantaneously and at $t=0$ we assume that the inductor has an initial current I_0

$$i(0) = I_0$$

with the corresponding value of the energy stored as

$$w(0) = \frac{1}{2} L I_0^2$$

Applying KVL around the loop

$$v_L + v_R = 0$$

The Source-Free RL Circuit

Applying KVL a to the loop,

$$v_L + v_R = 0$$

By definition, $v_L = L \, di/dt$ and $v_R = iR$. Thus,

$$L \frac{di}{dt} + Ri = 0$$

$$\frac{di}{dt} + \frac{R}{L}i = 0$$

This is a first-order differential equation,

Rearranging terms and integrating gives

$$\int_{I_0}^{i(t)} \frac{di}{i} = - \int_0^t \frac{R}{L} dt$$

$$\ln i \Big|_{I_0}^{i(t)} = - \frac{Rt}{L} \Big|_0^t \quad \Rightarrow \quad \ln i(t) - \ln I_0 = - \frac{Rt}{L} + 0$$

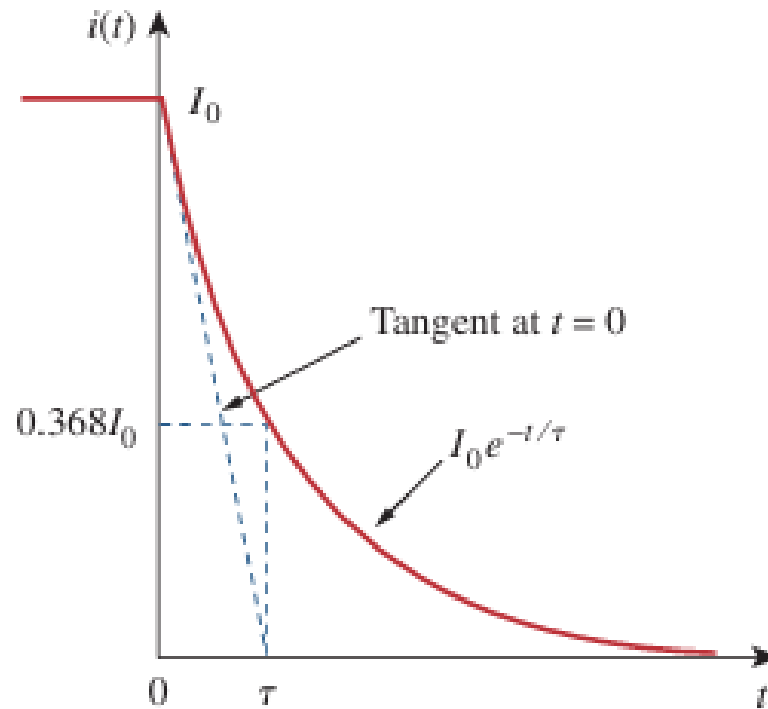
$$\ln \frac{i(t)}{I_0} = - \frac{Rt}{L}$$

The Source-Free RL Circuit

Taking powers of e produces,

$$i(t) = I_0 e^{-Rt/L}$$

The equation shows that the natural response of the RL circuit is an exponential decay of the initial current.



The natural response of a circuit refers to the behavior (in terms of voltages and currents) of the circuit itself, with no external sources of excitation.

The Source-Free RL Circuit

The time constant for the RL circuit is

$$\tau = \frac{L}{R}$$

Thus,

$$i(t) = I_0 e^{-t/\tau}$$

τ has the units of Seconds. The smaller the time constant of a circuit, the faster the rate of decay of the response. The larger the time constant, the slower the rate of decay of the response.

Voltage across the resistor is $v_R(t) = iR = I_0 R e^{-t/\tau}$

The power dissipated in the resistor is $p = v_R i = I_0^2 R e^{-2t/\tau}$

The energy absorbed by the resistor is

$$w_R(t) = \int_0^t p(\lambda) d\lambda = \int_0^t I_0^2 R e^{-2\lambda/\tau} d\lambda = -\frac{\tau}{2} I_0^2 R e^{-2\lambda/\tau} \bigg|_0^t, \quad \tau = \frac{L}{R}$$

$$w_R(t) = \frac{1}{2} L I_0^2 (1 - e^{-2t/\tau})$$

The Source-Free RL Circuit

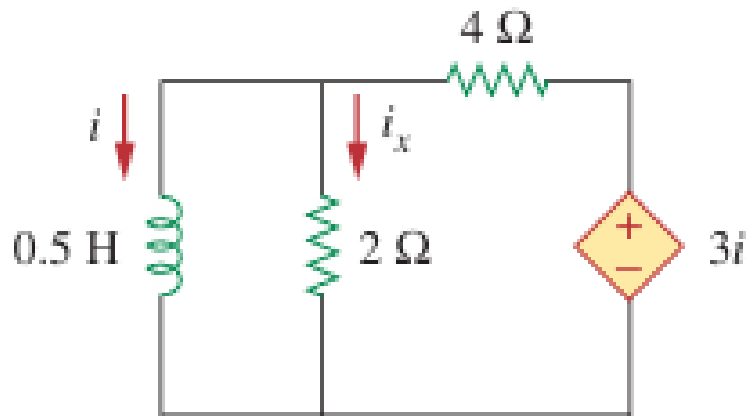
Notice that as t tends to infinity, w tends to $\frac{1}{2} L I_0^2$, which is the same as the energy initially stored in the inductor.

The energy that was initially stored in the inductor is eventually dissipated in the resistor.

When a circuit has a single inductor and several resistors and dependent sources, the Thevenin equivalent can be found at the terminals of the inductor to form a simple RL circuit. Also, one can use Thevenin's theorem when several inductors can be combined to form a single equivalent inductor.

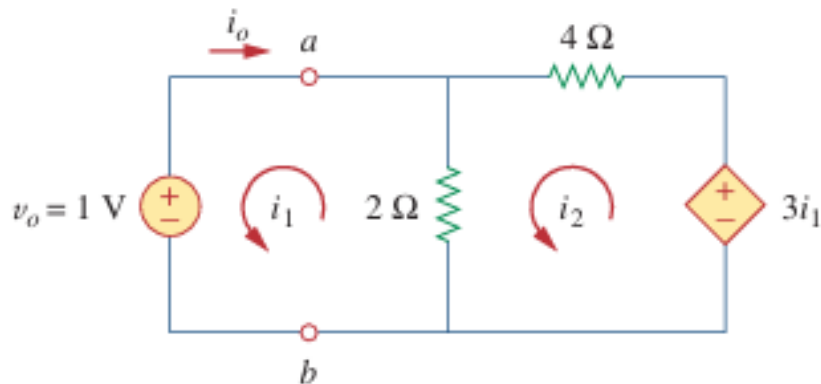
Exercise

1. Assuming $i(0) = 10$ A, calculate $i(t)$, and $i_x(t)$ in the circuit.



Because of the dependent source, we insert a voltage source with $v_o = 1$ V at the inductor terminals a-b.

Applying KVL to the two loops,



$$2(i_1 - i_2) + 1 = 0 \quad \Rightarrow \quad i_1 - i_2 = -\frac{1}{2}$$

$$6i_2 - 2i_1 - 3i_1 = 0 \quad \Rightarrow \quad i_2 = \frac{5}{6}i_1$$

$$i_1 = -3 \text{ A}, \quad i_o = -i_1 = 3 \text{ A}$$

Hence, $R_{\text{eq}} = R_{\text{Th}} = \frac{v_o}{i_o} = \frac{1}{3} \Omega$

Exercise

The time constant is

$$\tau = \frac{L}{R_{\text{eq}}} = \frac{\frac{1}{2}}{\frac{1}{3}} = \frac{3}{2} \text{ s}$$

Thus, the current through the inductor is

$$i(t) = i(0)e^{-t/\tau} = 10e^{-(2/3)t} \text{ A}, \quad t > 0$$

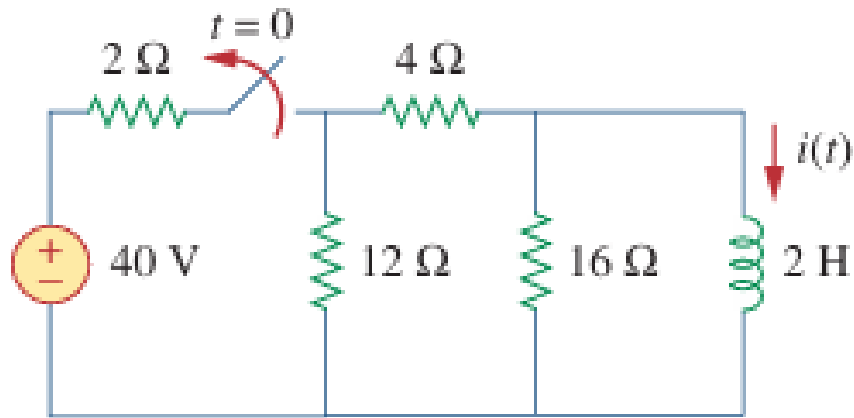
The voltage across the inductor is, $v = L \frac{di}{dt} = 0.5(10) \left(-\frac{2}{3} \right) e^{-(2/3)t} = -\frac{10}{3} e^{-(2/3)t} \text{ V}$

Since the inductor and the resistor are in parallel,

$$i_x(t) = \frac{v}{2} = -1.6667 e^{-(2/3)t} \text{ A}, \quad t > 0$$

Exercise

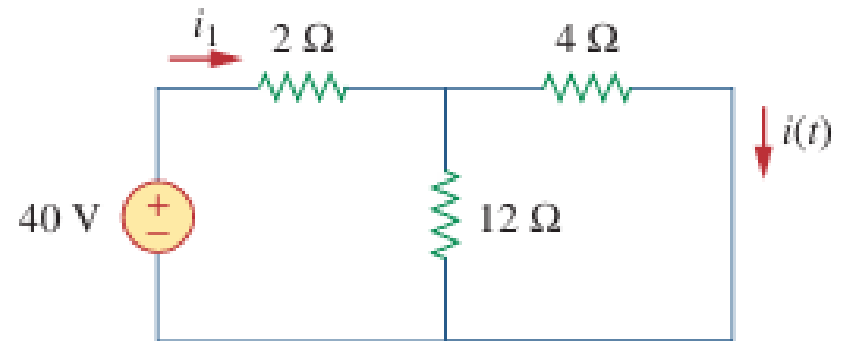
2. The switch in the circuit has been closed for a long time. At the $t = 0$, the switch is opened. Calculate $i(t)$ for t greater than zero.



$$\frac{4 \times 12}{4 + 12} = 3\Omega$$

$$i_1 = \frac{40}{2 + 3} = 8\text{ A}$$

When the switch is closed, and the inductor acts as a short circuit to dc. The 16Ω resistor is short-circuited;



We obtain $i(t)$ using current division

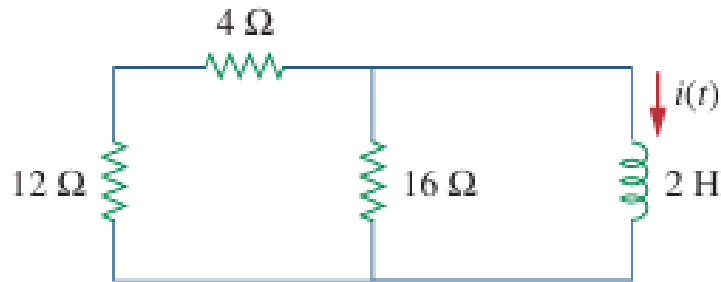
$$i(t) = \frac{12}{12 + 4} i_1 = 6\text{ A}, \quad t < 0$$

Exercise

Since the current through an inductor cannot change instantaneously.

$$i(0) = i(0^-) = 6 \text{ A}$$

When the switch is open at t greater than 0 and the voltage source is disconnected. We now have the source-free RL circuit,



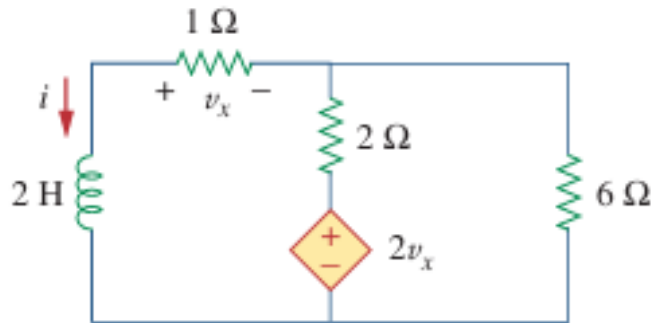
$$R_{\text{eq}} = (12 + 4) \parallel 16 = 8 \Omega$$

$$\tau = \frac{L}{R_{\text{eq}}} = \frac{2}{8} = \frac{1}{4} \text{ s}$$

$$i(t) = i(0)e^{-t/\tau} = 6e^{-4t} \text{ A}$$

Exercise

3. Find I and v_x in the circuit. Let $i(0) = 12A$



Next class: Singularity functions